

One-Class SVM for PFAS–MOF Descriptor Modeling

Problem framing

- **Goal:** Learn the *region* of descriptor space occupied by known PFAS-adsorbing MOFs (positive-only data).
 - **Why OC-SVM:** No reliable negatives or scalar target; need a soft boundary around positives.
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Descriptor set

Core (chemist-prioritized): - PLD (Å) - LCD (Å) - Geometric Surface Area (m²/g) - Void fraction - Pore volume (cm³/g)

Candidate / exploratory: - Density (g/cm³)

Dimensionality: 5–6 features is appropriate for OC-SVM with ~36 samples.

Modeling choices

- **Single multivariate OC-SVM** is standard; >2 features is fine (2D examples are for visualization only).
 - **Kernel:** RBF (default for nonlinear boundaries in materials space).
 - **ν (nu):** Controls allowed outliers (e.g., 0.1–0.2 as a starting range).
 - **γ (gamma):** Set via median pairwise distance heuristic or limited grid search.
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Critical preprocessing

- **Scaling is mandatory.** Use z-score or robust scaling.
 - Rationale: OC BF kernel relies on Euclidean distances; unscaled features (e.g., GSA) will dominate.
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When to train multiple models

Not required for dimensionality, but useful for **scientific interrogation**: - Geometry-only vs porosity-only vs all-features models - All-features vs all + density

Purpose: test stability and hypothesis sensitivity, not ensembling.

Assessing feature importance (geometric, not parametric)

There is no native coefficient-based importance. Use **sensitivity analyses**:

1) Leave-One-Feature-Out (LOFO)

- Retrain after removing each feature.
- Compare:
 - Fraction of training points flagged as outliers
 - Changes in anomaly score distribution
 - Boundary stability

2) Perturbation sensitivity

- Perturb one feature ($\pm 5\text{--}10\%$) and observe score changes.
- Interpretable for domain experts.

3) Linear OC-SVM (diagnostic only)

- Fit a linear OC-SVM to inspect weight magnitudes/signs.
- Use for sanity checks; do not over-interpret physically.

4) Explainability tools (optional)

- SHAP on the decision function/anomaly score (qualitative only; small-n caution).
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What not to do

- Do not report p-values or regression-style coefficients.
 - Do not assume monotonic effects.
 - Do not over-tune to force perceived importance.
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Recommended workflow

1. Train RBF OC-SVM on core features (scaled).
 2. Validate stability (v , γ sensitivity).
 3. Add density; compare boundary and scores.
 4. Perform LOFO and perturbation analyses.
 5. Report *boundary stability and sensitivity*, not coefficients.
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Outputs to report

- Anomaly scores (continuous)

- In/out decision at chosen threshold
- Sensitivity results supporting descriptor relevance